

1 **Development of a Simple R Package (aLBI) for the Estimation of Stock Status from the**
2 **Length Frequency Data**

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ABSTRACT

Stock assessment is a difficult task for sustainable fisheries management especially when population and fisheries data are limited. This study aims to develop a simple and well-tested R package (aLBI) to facilitate the fish stock status assessment by providing robust functions for estimating length-based indicators (LBIs) and the stock health by utilizing the length frequency data. The primary objective is to streamline the calculation of different size classes and visualization of the estimated metrics for more accurate and efficient understanding. The package includes two main functions for calculating key LBIs such as maximum length (L_{max}), asymptotic length (L_{∞}), length at sexual maturity (L_{mat}), optimum length (L_{opt}), $L_{opt} \pm 10\%$ (L_{opt_p10} , L_{opt_m10}) and percentages of fish at maturity (P_{mat}), optimum length (P_{opt}), and mega spawners (P_{mega}) with confidence intervals by bootstrapping method. This study also evaluated the performance of accuracy of this package by validating with real world length frequency data of *Sperata aor* with different bin size. The findings with a calculated optimum bin size three demonstrate its utility in producing reliable stock assessments, with mean estimates for L_{max} , L_{∞} , L_{mat} , and L_{opt} at 74.27 cm, 78.17 cm, 41.83 cm, and 44.77 cm, respectively, as well as respective confidence intervals. Besides, percentages of fish at maturity, optimum length, and mega spawners are 12.48%, 9.03%, and 5.09%, against targets of 100%, 100%, and 20% respectively. Additionally, the stock status indicators show probabilities of 67% for both TSB40 and LSB25 for base model. By integrating advanced statistical methods and user-friendly interfaces, aLBI enables effective monitoring of fish populations and informed decision-making. Tested on various bin size (1, 2, 4 and 5) with a dataset, the package consistently performs accurately. Its ease of use and robust inferences make it an essential resource for researchers and managers. Future recommendations include expanding functionality to cover more species and integrating additional biological parameters for enhanced stock assessments. Continued collaboration with the fisheries science community will ensure aLBI remains a cutting-edge tool for sustainable fisheries practices.

Key words: R package development, stock assessment, aLBI, length-based indicators (LBIs), target biomass, limit biomass, Kaptai Lake.

51 1 INTRODUCTION

52 Fish stock assessment is a crucial element in the management and conservation of fishery
53 resources, aiming to provide reliable estimates of fish population dynamics, exploitation
54 levels, and sustainable yield. Effective fish stock assessment ensures that fishery management
55 decisions are based on sound scientific evidence, promoting the long-term sustainability of
56 marine ecosystems and the fishing industry (Hilborn & Walters, 1992; FAO, 2020). It allows
57 for the determination of fishing quotas, the identification of overfished stocks, and the
58 implementation of conservation measures. Without reliable stock assessments, fisheries
59 management cannot effectively balance exploitation with conservation, leading to the
60 depletion of fish stocks and the collapse of fisheries (Pilling et al., 2008). Thus, knowledge
61 about the population status is critical for effective management (Alam et al., 2021). However,
62 the application of traditional stock assessment methods to these populations is often hampered
63 by numerous changes, such as challenges in data collection due to the lack of monitoring and
64 control programs (Hilborn and Walters, 1992; Scott et al., 2016). Given this situation, several
65 stock assessment methods have been developed for data-limited fisheries. Among these
66 methods, some require yearly landing data and life history parameters (i.e., natural mortality,
67 maximum age, age at maturity) (Martell and Froese, 2013; Thorson and Cope, 2015; Zhou et
68 al., 2017; Free et al., 2017; Canales et al., 2018), while other require length-frequency data in
69 cohort analysis models (e.g., Hordyk et al., 2014; Froese et al., 2018; Prince et al., 2015; Rudd
70 and Thorson, 2018; Ault et al., 2019). Collecting length-frequency data is often more
71 straightforward and cost-effective than gathering of landing landings data, and the resultant
72 data tend to be more reliable (Chong et al., 2020; Pilling et al., 2008; Hordyk et al., 2015;
73 Mildenberger et al., 2017; Froese et al., 2018).

74 Thus, length-based stock evaluation methods provide a good and typically more practical
75 option (Pauly & Morgan, 1987). Using length frequency data, these techniques estimate
76 significant population parameters like patterns of recruitment, growth rates, mortality rates
77 due from natural causes and fishing, and so on (Sparre & Venema, 1998). One consistent
78 approach to find length frequency information is to look at fish landings or market samples.
79 (ref). Due to their economic viability and readily accessibility application of length frequency
80 data based stock assessment models become increasingly popular, particularly in areas with
81 low financial and technological resources (Gayanilo & Sparre et al., 2005; Hoggarth et al.,
82 2006).

83 The *Sperata aor* (Hamilton, 1822), commonly known as the long-whiskered catfish, is an
84 important species within the Bagridae family, widely distributed across freshwater systems

85 in South Asia, including the river basins of the Ganges, Brahmaputra, and Indus. In
86 Bangladesh, this species is notably prevalent in Kaptai Lake, an artificial reservoir created by
87 damming the Karnaphuli River in the Chittagong Hill Tracts in 1966. Kaptai Lake is vital for
88 supporting local fisheries, including *Sperata aor*, which plays a crucial role in the livelihoods
89 of numerous fishing communities in the region (Hossain et al., 2012). The lake is a significant
90 source of fish for local markets, contributing to both food security and the local economy
91 (Shalehin et al., 2022). However, increased fishing pressure, along with environmental
92 changes and habitat degradation, has raised concerns about the sustainability of many fish
93 stocks including *Sperata aor* populations in Kaptai Lake (Khatun et al., 2023). This situation
94 highlights the need for effective management and conservation strategies to ensure the long-
95 term viability of this vital fishery. Given the cultural and economic significance of *Sperata*
96 *aor* in local diets and traditional practices, sustainable exploitation and conservation efforts
97 are crucial for maintaining ecological balance in Kaptai Lake and preserving the biodiversity
98 that supports local communities (Rana et al., 2024).

99 Numerous metrics based on length data have been developed to evaluate the status of fish
100 stocks. Among these, three length-based indicators (P_{mat} , P_{opt} , P_{mega}) proposed by Froese
101 (2004) have gained prominence due to their simplicity and utility. These indicators are
102 grounded in the principles of “let them spawn,” “let them grow,” and “let the mega spawners
103 live,” aiming to avoid growth and recruitment overfishing while managing exploitation
104 sustainably. The first indicator, P_{mat} quantifies the proportion of the catch composed of fish
105 that have reached sexual maturity, reflecting the reproductive potential. The second, P_{opt}
106 measures the percentage of fish within the optimal size range for harvest, representing
107 individuals which are large enough to have contributed to spawning but not so large as to
108 exhibit declining growth and fecundity. Lastly, P_{mega} tracts the proportion of very large, older
109 individuals in the catch which are critical for population resilience due to their high
110 reproductive potential and genetic diversity.

111 These indicators assess exploited fish stocks using straightforward, data-limited approaches
112 based on length-frequency distributions. They are particularly advantageous for fisheries
113 where detailed stock assessments are impractical, as they focus on the distribution of fish
114 lengths relative to key biological reference points (Froese, 2004; Cope & Punt, 2009). By
115 simplifying the evaluation of stock sustainability, these metrics serve as practical tools for
116 both scientific research and fisheries management in data-limited settings. Given that the
117 simplicity and length data requirement, this study aimed to evaluate the stock status of
118 *Sperata aor* in the Kaptai Lake. Furthermore, this study also primarily aims to create a

scientifically valid, user-friendly, robust R software for length-based fish stock assessment. Thus, this study intends to design and develop an R package that integrates advanced statistical models and visualization tools for fisheries data analysis. In addition, it aims to provide comprehensive documentation and user support for the package and validate the R package by using real-world length frequency data of *Sperata aor*.

2 METHODOLOGY

2.1 Study area

This study focuses on the collection of length frequency data of *Sperata aor* from Kaptai Lake, located in the Chattogram district of Bangladesh (latitude: 22°30'00" N and longitude: 92°13'00" E). The lake lies in the southeast part of Bangladesh, within the coordinates of latitude 22°22' to 23°18'N and longitude 92°00' to 92°26'E (Figure 1). Kaptai Lake, the largest man-made freshwater lake in the country with a water area of 68,000 hectares which is 1.78% of the total inland open water bodies (DoF, 2022), is a vital resource for local fisheries and communities. The lake interconnects several upazilas, including Rangamati Sadar, Langadu, Nannerchar, Jurachari, Kaptai, and Belaichari. It is fed by the Karnaphuli, Chengi, Mayani, Kassalong, and Rinkhyong rivers. Fishing activities in Kaptai Lake play a crucial role in the livelihoods of local communities and contribute significantly to Bangladesh's inland fisheries production. As the largest artificial lake in the country, Kaptai Lake provides an expansive fishing ground, supporting both commercial and subsistence fishing practices. The lake is home to a wide variety of freshwater fish species, including native species like catfish (*Sperata aor*, *Mystus spp.*) and carps (*Labeo rohita*, *Cirrhinus mrigala*), as well as introduced species like tilapia (*Oreochromis spp.*).

The Kaptai lake fishery is a multi-species and multi-gear fisheries where fishing methods employed in range from traditional techniques such as cast nets, gill nets, traps, and seine nets to modern fishing gear for commercial operations. Seasonal fishing activities, heavily influenced by monsoon patterns, are vital for meeting the local demand for protein and generating income. However, overfishing, habitat degradation, and water level fluctuations pose challenges to the sustainability of these activities. Efforts to manage the lake's fisheries include community-based approaches and government interventions aimed at ensuring stock sustainability, such as seasonal fishing bans and the introduction of conservation zones.

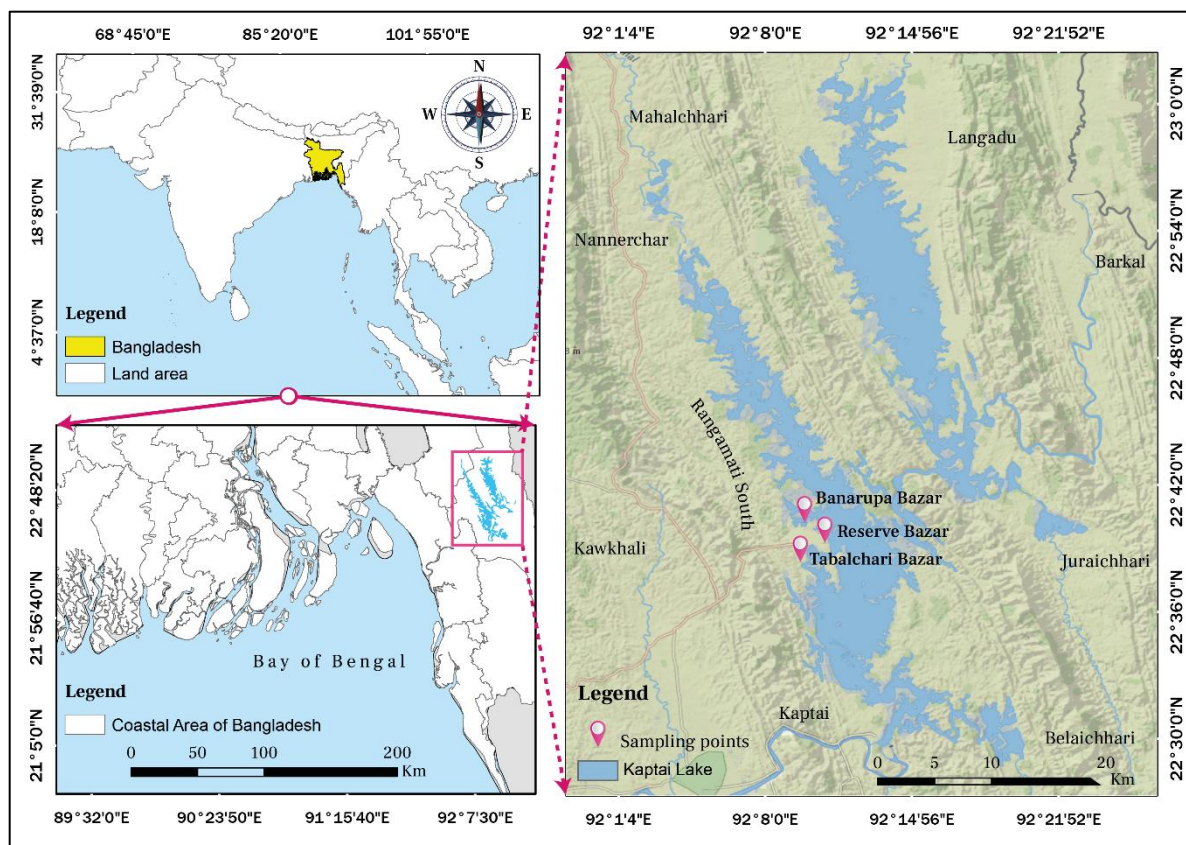


Figure 1: Map showing the sampling points of the study

2.2 Collection of length frequency data

Length-frequency data were collected from four sampling stations: the Rangamati and Kaptai landing centers, along with two local markets, Banarupa Bazar and Reserve Bazar. Sampling was conducted monthly from August 2022 to March 2023. However, no data were collected between April and July due to a fishing ban in this period. During months of high landings, a stratified random sample comprising 10% of well-mixed individuals of both sexes from each pile was measured. Conversely, during periods of low landings, all available individuals were sampled. At the landing centers, fish lengths were measured to the nearest 0.1 cm using a measuring scale.

Sampling Calendar												
Months Years	January	February	March	April	May	June	July	August	September	October	November	December
2022												
2023												

Figure 2: Sampling calendar of the study

2.3 Data analysis

2.3.1 Main functions of aLBI package

Table 1. Main Functions of aLBI R package

Functions	Descriptions	Usage	Arguments
FrequencyTable	This function creates a frequency distribution table for fish length data	FrequencyTable(data, method, bin_width, Lmax)	data: A numeric vector of fish lengths. bin_width: Optional. A numeric value specifying the bin width for class intervals, if left NULL, the bin width is automatically calculated using Wang et al. (2020) optimum bin size (OBS) formula. Lmax: Optional. The maximum length of fish. If left NULL then automatically calculate the frequency table with the observed maximum length of the provided data.
FishPar	Calculates various length-based indicators for fish stock assessment using length frequency data and bootstrap resampling.	FishPar(data, resample, progress)	data: A data frame containing two columns: Length and Frequency. resample: An integer indicating the number of bootstraps resamples. progress: A logical value indicating whether to display progress.
FishSS	Assesses the stock status based on parameters calculated by the FishSS function	FishSS(data, LM_ratio, P_obj, P_mat, P_opt)	data: A data frame containing the necessary columns for stock status calculation. (Cope & Punt, 2009) LM_ratio: A numeric value representing the ratio of L_{mat} and L_{opt}. P_obj: A numeric value representing the sum of P_{mat}, P_{opt} and P_{mega}. P_mat: A numeric value representing the percentage of mature fish. P_opt: A numeric value representing the percentage of optimally sized fish.

The FrequencyTable function generates the frequency table as well as provide a data frame containing the upper length class and their corresponding frequencies. With this frequency table the FishPar function facilitates the analysis by estimating key biological parameters, such as the maximum length (L_{max}), asymptotic length (L_{∞}), and length at maturity (L_{mat}),

using bootstrap resampling techniques (Table 1). It ensures data integrity by validating input data and generating robust confidence intervals for parameter estimates. Additionally, it provides insightful visualizations to help interpret the results, making it a valuable tool for researchers and fishery managers. In addition, the FishPar function also calculate particular indicators such as L_{mat}/L_{opt} ratio (LM_ratio) and sum of P_{mat} P_{opt} P_{mega} (P_{obj}) to ascertain the stock situation. Analyzing these indications helps one to choose suitable columns from the input data to create important stock metrics such Limit Spawning Biomass (LSB25) and Target Spawning Biomass (TSB40) (Table 1). This dual-function strategy improves our knowledge of fish population dynamics, therefore helping to guide the development of sensible management plans and fisheries policies.

2.3.2 Calculating frequency table

Frequency distribution table was constructed using the following formula of optimum bin size (OBS) recommended by Wang et al. (2020), facilitate by the *FrequencyTable* function of aLBI package.

$$\text{Optimum Bin Size (OBS)} = (0.23 * (L_{max}^{0.6})) \dots \dots \dots (1)$$

L_{max} = Maximum length

2.3.3 Estimating growth parameters by FishPar function

L_{max} will be estimated from the length frequency data by FishPar function as followed:

$$L_{max} = \max (data[1]) \dots \dots \dots (3)$$

Here, data[1] = first column of the length frequency data where length of the sampled data was recorded and pick up the largest length by max formula.

Then an initial value for asymptotic length (L_{∞}) was calculated using the following formula proposed by Pauly (1984):

$$L_{\infty} = \frac{L_{max}}{0.95} \dots \dots \dots (4)$$

To estimate the length at sexual maturity (L_m), an empirical equation proposed by Froese & Binohlan (2000) was used as followed:

$$\log(L_m) = 0.8979(\log(L_{\infty})) - 0.0782 \dots \dots \dots (5)$$

2.3.4 Estimating Length-Based Indicators (LBIs) by FishPar function

In order to evaluate the stock status in relation to exploitation and prevent excessive fishing that could harm development and recruitment, Froese (2004) suggested three straightforward length-based indicators (LBIs) known as P_{mat} , P_{opt} , and P_{mega} .

I. The first indicator P_{mat} , can be defined as the proportion of mature fish in the catch, and it represents the concept of “let them spawn!” The objective is to ensure that every single fish (100%) is able to reproduce at least once before being caught, in order to restore and sustain a robust spawning biomass of fisheries. The percentage of fish in the capture that have a length larger than the length at sexual maturity (L_m) was determined using the following formula proposed by Froese (2004):

$$P_{mat} = \sum_{L_{mat}}^{L_{max}} P_L \dots \dots \dots (6)$$

Where, P_L is the percentage of fish in the catch composition with a length interval of L .

II. Indicator two P_{opt} , defined as the percentage of fish caught at their optimal length, which is the length that maximizes the yield and revenue. The optimum length is generally somewhat bigger than the length at initial maturity and can be readily determined using growth and mortality parameters or empirical equations (Froese & Binohlan, 2000). The objective is to capture every fish (100%) within a range of around $\pm 10\%$ (between $0.9L_{opt}$ and $1.1L_{opt}$) of the optimal length. L_{opt} and P_{opt} were calculated by following formula:

$$\log(L_{opt}) = 1.053(\log(L_{mat})) - 0.0565 \dots \dots \dots (7)$$

III. The P_{mega} , defined as the percentage of old and larger fish (mega = mega spawner) in the catch composition. The metric is calculated as the proportion of mature fish in the catch that exceed the optimal length by 10%. The purpose of the management regime is to execute a fishing technique that ensures no mega-spawners are caught, with a goal of 0% catch rate. In the absence of such strategy, a range of 30-40% mega-spawners is indicative of a robust structure and is considered favorable (Froese, 2004). However, if the proportion falls below 20%, it is a cause for concern. The following equation was used to estimate P_{mega} proposed by Froese (2004):

$$P_{mega} = \sum_{1.1L_{opt}}^{L_{max}} P_L \dots \dots \dots (8)$$

231 This FishPar function also calculate the length at sexual maturity ()

232 2.3.5 Stock spawning biomass (SB) evaluation

233 FishSS function evaluate the stock status (limit spawning biomass and target spawning
234 biomass) from a decision table (*CPdata*) (Figure 3) demonstrated by Cope and Punt (2009)
235 which is readily available with the aLBI package based on the input parameters (*LM_ratio*,
236 *Pmat*, *Popt*, *Pmega*) estimated by FishPar function. For detail understanding about the
237 following flowchart see the *CPdata* provided by the aLBI package.

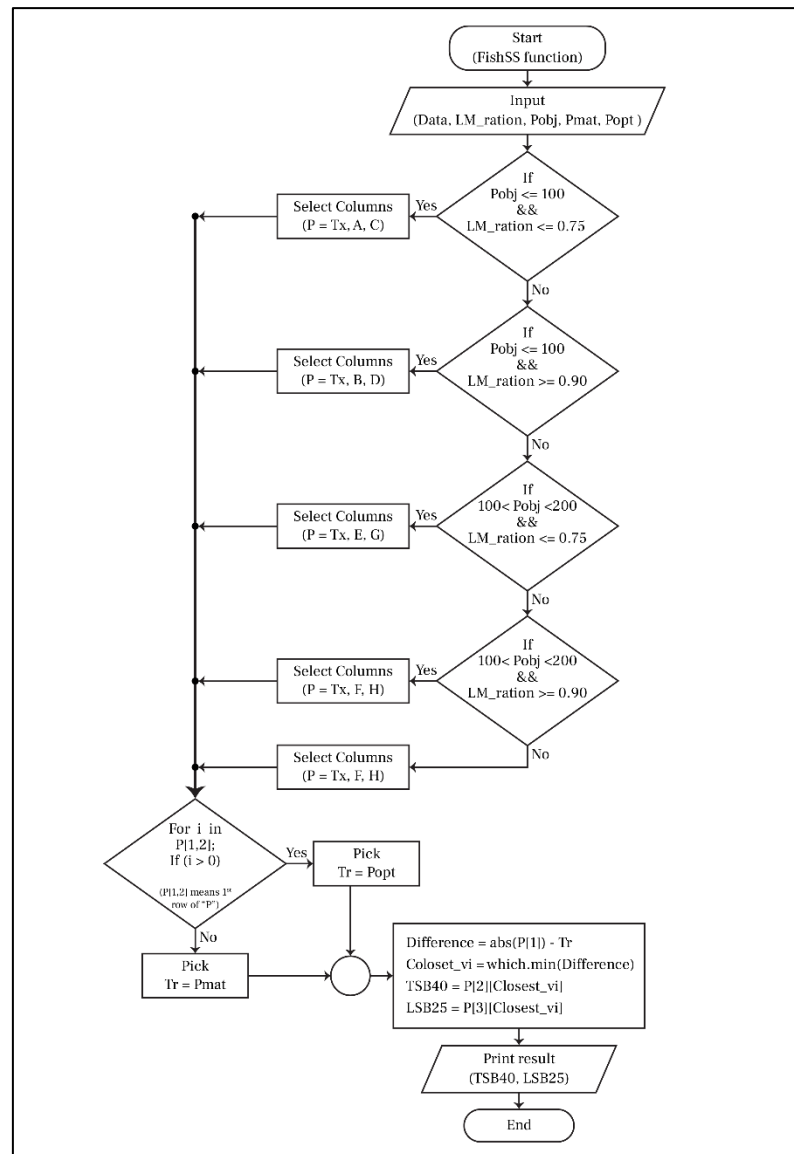


Figure 3: Flow chart of the decision tree adopted from Cope and Punt (2009)

241 2.3.6 Evaluating the robustness of the aLBI package

242 To evaluate the robustness of the aLBI package under varying bin sizes, the length frequency
243 data were rearranged using the *FrequencyTable* function of the aLBI package with bin sizes

of 1, 2, 4, and 5, in addition to the base model with a bin size of 3. The uncertainty in length-based indicators (LBIs) (P_{mat} , P_{opt} , and P_{mega}) was propagated from the uncertainty in key length parameters such as length at maturity (L_{mat}), optimal length (L_{opt}), and maximum length (L_{max}). The uncertainty in these length parameters was estimated through a bootstrapping approach, which resampled the length data 1,000 times to generate confidence intervals for each parameter. The bootstrapped estimates were then used to calculate the LBIs for each bin size. This approach ensured that the variability inherent in the length data was accounted for in the robustness evaluation. Statistical analyses, including one-way ANOVA and Tukey's HSD test, were performed to test for significant difference in length parameters as well as LBI estimates across the different bin sizes.

3 RESULTS

3.1 Installation of the package

The package (aLBI) is accessible on GitHub repository (for latest update) and also available on CRAN (<https://cran.r-project.org/web/packages/aLBI/index.html>). For detailed installation instructions, troubleshooting tips, and additional information, users can refer to the vignettes (<https://cran.r-project.org/web/packages/aLBI/vignettes/Introduction.html>).

3.2 Observed output of FishPar function

Length frequency data of *Sperata aor* was given input to the aLBI package for estimating the parameters and visualization of the plots for understanding the stock status.

3.2.1 Frequency distribution plot

The total lengths (TL) of 1177 *Sperata aor* individuals, ranging from 9.5 to 75 cm were measured. A frequency table with an optimum class intervals of 3 cm and a unimodal length frequency distribution plot (Figure 4) was produced by aLBI package. The observed frequencies represented by bars, reveals the highest abundance of individuals within the length class of 27 cm (156 individuals) and followed by 30 cm (143 individuals), 21 cm (137 individuals), 24 cm, 18cm both with the same count (135 individuals), indicating these lengths are common among the sampled fish population.

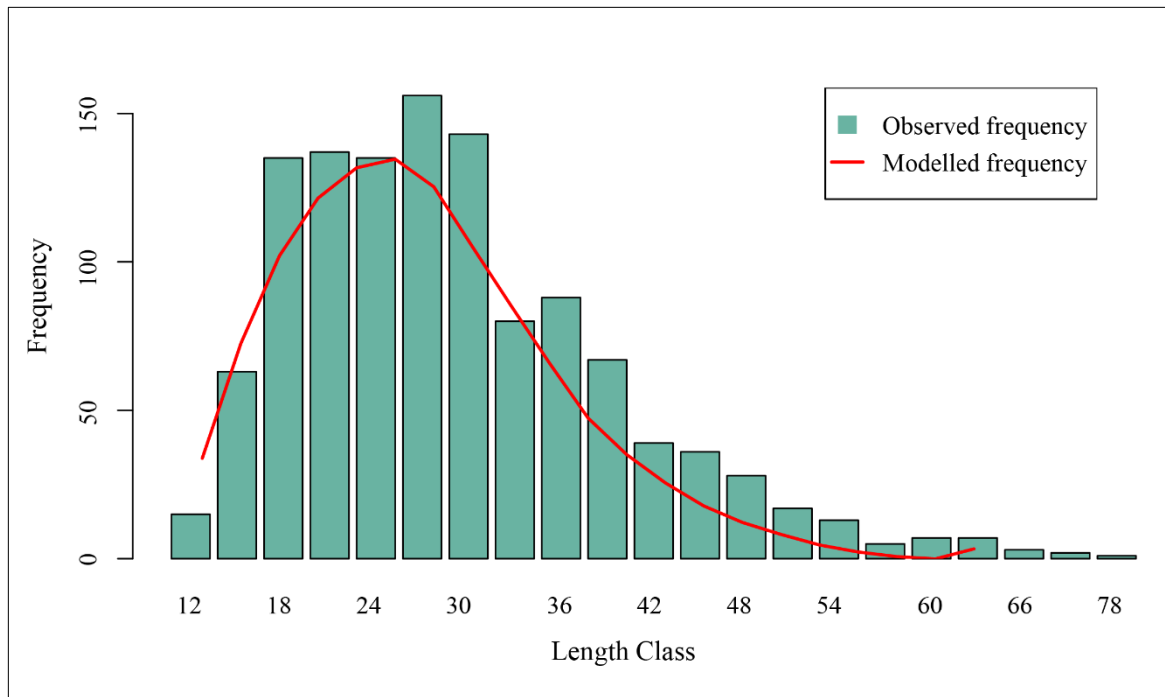


Figure 4: Plot showing the length frequency distribution of *Sperata aor*.

The red curve, representing the model frequency, aligns closely with the observed data and highlighting the accuracy of the applied model. The plot was found to be heavily skewed towards left side, indicating good recruitment but also potential growth overfishing.

3.2.2 Estimating different length groups

The aLBI package produced a plot (Figure 5) for different length groups of *Sperata aor*. The length-based metrics provided a comprehensive understanding of the fish stock status. The plot depicts the mean maximum length (L_{max}) of 74.27 cm with a confidence interval (CI) 63.00 to 78 cm and asymptotic length (L_{∞}) of 78.17 cm with a CI of 66.32 to 82.11 cm, indicating that the species has the potential to attain considerable sizes under optimal conditions and minimal fishing pressure. The mean length at sexual maturity (L_{mat}) of 41.83 cm (ranging from 36.09 to 43.72 cm). The mean optimum length (L_{opt}) is 44.77 cm. On the contrary, the estimated range for $L_{mat} \pm 10\%$ of L_{opt} is = 40.29 to 49.24 cm, represents the ideal range which produces maximum biological yield.

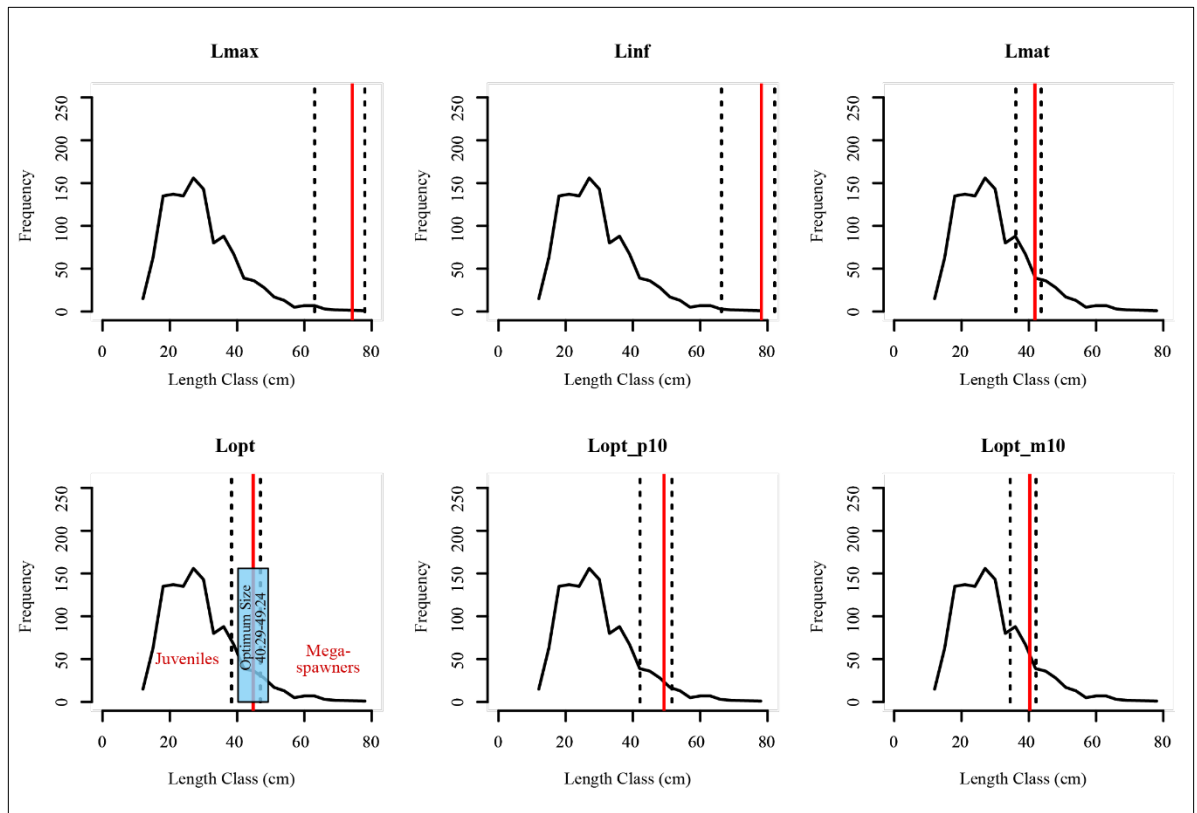


Figure 5: Plot showing the frequency line with different length parameters of *Sperata aur*. Solid red lines depict the mean estimates and dotted lines depicts the lower (LCI) and upper confidence intervals (UCI).

Estimated length based parameters of *Sperata aur* and their respective confidence intervals are showing in the Table 2.

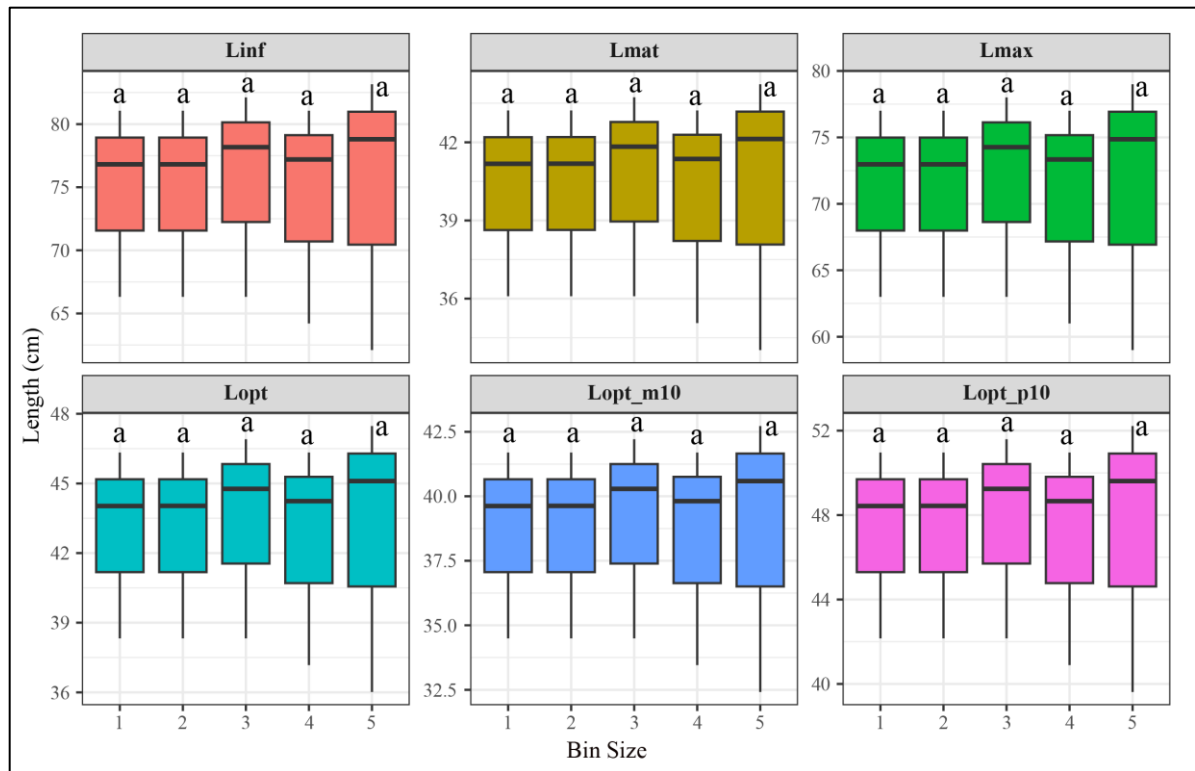
Table 2: Different length parameters of *Sperata aur*

Parameters	Mean (cm)	LCI	UCI
L_{∞}	78.17	66.32	82.11
L_{max}	74.27	63.00	78.00
L_{mat}	41.83	36.09	43.72
L_{opt}	44.77	38.32	46.90
L_{opt_p10}	49.24	42.16	51.59
L_{opt_m10}	40.29	34.49	42.21

3.3 Scenario produced with different bin size for length parameters

The boxplot (Figure 6) illustrates the consistency of the estimated length parameters (L_{∞} , L_{mat} , L_{max} , L_{opt} , L_{opt_m10} , and L_{opt_p10}) across different bin sizes (1 to 5, with the base model bin size of 3). The estimated values of L_{∞} and L_{max} exhibited minimal deviations from the base model, ranging from +1.74% to -0.8%, while L_{mat} and L_{opt} showed deviations of +1.57% to -0.71%

301 and +1.65% to -0.75%, respectively. Statistical analysis using One-Way ANOVA and
 302 Tukey's post-hoc test confirmed that the differences in estimated length parameters across
 303 different bin sizes were not significant ($p > 0.05$).



304
 305 **Figure 6: Distribution of length parameters with different bin size (base model bin size**
 306 **= 3), represented with the mean, lower and upper confidence intervals. Shared letter (a)**
 307 **indicating no significant differences ($p > 0.05$) in the estimation of length parameters**
 308 **with different bin size based on one-way ANOVA and Tukey's HSD.**

309 3.3.1 Length-Based Indicators (LBIs)

310 The simple set of indicators (P_{mat} , P_{opt} , P_{mega}) proposed by Froese (2004), were estimated for
 311 the stock of *Sperate aor* (Table 3) reveal significant deviations from the target reference
 312 values. The percentage of sexually mature fish (P_{mat}) was 12.48% (7.55–22.69%), far below
 313 the target of 100%, indicating heavy exploitation of immature fish and reduced reproductive
 314 potential. Similarly, the percentage of optimally sized fish (P_{opt}) was 9.03% (5.14–17.24%),
 315 substantially below the 100% target, reflecting insufficient numbers of fish reaching their full
 316 growth potential. The proportion of mega-spawners (P_{mega}), crucial for genetic diversity and
 317 population resilience, was only 5.09% (2.41–11.79%), compared to the target of 20%,
 318 highlighting that this fishery has already removed major portion of its larger individuals. The
 319

combined indicator (P_{obj}) was 26.61% (15.09–51.71%), further underscoring the stock's unsustainable status.

Table 3: Estimated length-based indicators (LBIs)

Parameters	Mean (%)	LCI	UCI
P_{mat}	12.48	7.55	22.69
P_{opt}	9.03	5.14	17.24
P_{mega}	5.09	2.41	11.79
P_{obj}	26.61	15.09	51.71

3.3.2 Scenario produced with different biz size for the aLBI indicators

The boxplot (Figure 8) illustrates the robustness of the method by producing identical results for (P_{mat} , P_{opt} , P_{mega}) across different biz sizes (1 to 5, with a base model bin size of 3) using aLBI package. The propagated uncertainty from the length parameters (L_{mat} , L_{opt} and L_{max}) resulted in minor deviations from the base model. Specifically, the deviation for P_{mat} ranges from +0.13% to -2.60%, for P_{opt} -0.4% to -3.25% and for P_{mega} -1.07% to 1.53%. Statistical analysis using One-Way ANOVA and Tukey's post-hoc test confirmed that the differences in estimated length parameters across different bin sizes were not significant ($p > 0.05$).

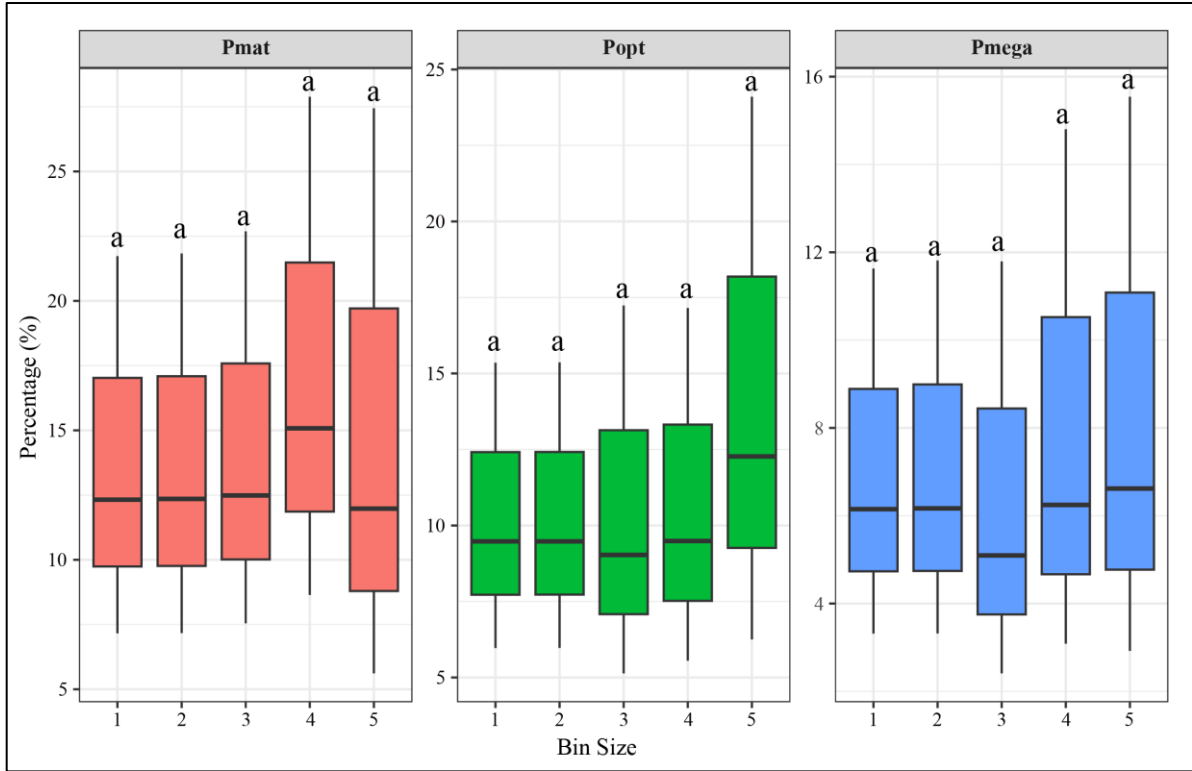


Figure 8: Distribution of LBIs with different bin size, represented with the mean, lower and upper confidence intervals. Shared letter (a) indicating no significant differences (p

> 0.05) in the estimation of length parameters with different bin size based on one-way ANOVA and Tukey's HSD.

3.4 Observed stock status of FishSS function

FishSS function facilitate the evaluation of stock status by quantifying the decision tree process. The sum of the all the three proportions (P_{mat} , P_{opt} , P_{mega}), termed as P_{obj} , provides comprehensive information to distinguish the selectivity patterns. However, the findings (Table 4) suggest that there is a probability that the stock biomass below the target reference points or exceed both the target biomass level ($TSB40$) and the limit biomass level ($LSB25$). These reference points were also found below the target reference points with different bin size.

Table 4. Estimated stock status of the *Sperata aor* by aLBI package

Bin size	Probability (%) of TSB40	Probability (%) of LSB25
1	67	67
2	67	67
3	67	67
4	52	44
5	67	67

4 DISCUSSION

4.1 Development and application of the aLBI package

Especially in data-limited conditions, the development of the aLBI package was motivated by the need of robust tools to evaluate fish stock situation using length-based indicators. Applying methods proposed by Froese (2004) and Cope & Punt (2009), the package generates a whole framework for computing significant parameters like L_{max} , L_{∞} , L_{mat} , and L_{opt} as well as percentage indicators including P_{mat} , P_{opt} , and P_{mega} . These tools enable data-driven insights capable of guiding sustainable fisheries management. The ability of the program to visualize length frequency distributions and calculate confidence intervals helps managers and fisheries researchers to evaluate the health of any fish stock along with the uncertainties associate with the estimated parameters. By allowing managers to use effective conservation measures and reach sustainable exploitation rates, the package helps decision-making by means of clear, pragmatic indicators.

4.2 Performance evaluation of aLBI package

4.2.1 Frequency table and distribution curve

In addition to calculating the default optimum bin size using the formula proposed by Wang et al. (2020), the FrequencyTable function in the aLBI package allows users to specify a custom bin size. This feature significantly enhances the package's flexibility, enabling users to tailor the frequency table construction to their specific dataset or analytical needs. In this study, the optimum bin size was calculated to be 3.07, which was rounded to 3 for practical purposes and used as the base model bin size. Using this bin size, the FishPar function constructed the frequency distribution curve for *S. aor*. The curve demonstrates a gradual decline in frequency beyond the 30 cm length class, indicating a reduced number of individuals in larger size categories. The dominance with smaller-length individuals (Juveniles) indicates this fishery mainly targeting the juveniles. Mia et al. (2024) and Bashar et al. (2021) reported a similar selectivity pattern for *Ailia coila* and *Eutropiichthys vacha* in this Kaptai Lake. According to the “The Protection and Conservation of Fish Act, 1950”, catching fish with a length smaller than 23 cm is prohibited in this Lake (Rayhan et al., 2021). However, present and previous findings (Mia et al., 2024; Bashar et al., 2021) revealed the violation of this rules because of the absence of proper monitoring settings.

4.2.2 Different length parameters and their uncertainty

The length parameters such as L_{max} , L_{∞} , L_{mat} , and L_{opt} are important matrices for fisheries management. L_{max} serves as an indicator of the population structure and is closely associate with the fishing activities. The mean length at maturity (L_{mat}) underscores the critical threshold for reproductive capacity, while catching fish at the optimum length (L_{opt}) maximizes biological yield. These metrics emphasize the necessity of implementing size limits to protect both juveniles and mature fish. no prior studies have investigated these parameters for *S. aor* in Kaptai Lake.

Table 4: Estimated length parameters from different region of South Asia.

Regions	L_{∞}	L_{max}	L_{opt}	L_{mat}	Reference
Bangladesh	98.95	94.00	56.65	52.66	Rahman, 2005
India	125.26	119	70.80	65.08	Khan et al., 2019
India	192.42	182.8	106.24	95.68	Day, 1878; Misra, 1959
India	128.42	122.00	72.49	66.55	Job et al., 1955
India	166.44	158.12	92.63	84.00	Saigal, 1964
India	108.72	103.28	61.92	57.30	Ramakrishniah, 1992
India	89.89	85.40	51.74	48.31	Nazir & Khan, 2020
Pakistan	189.47	180.00	104.70	94.36	Talwar & Jhingran, 1991; Tripathi, 1996

	Bangladesh	78.17	74.27	44.77	41.83	Present study
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Previous studies conducted in various South Asian regions (e.g., Day, 1878; Misra, 1959; Talwar and Jhingran, 1991; Tripathi, 1955; Rahman, 2005; Sigal, 1964; Ramakrishniah, 1992; Khan et al., 2019; Job et al., 1955) have reported estimates for L_{max} , L_{∞} , L_{opt} , and L_{mat} that are significantly higher than the present study's findings (Table 4). While habitat quality and environmental conditions are critical factors influencing fish growth (Alam et al., 2012), intense fishing pressure (especially high removal of juveniles) plays a major role in reducing the mean length of fish populations (Alam et al., 2021). This reduction in mean length contributes to declines in key length-based parameters and drives earlier maturation, as fish are increasingly exploited before they reach their full growth potential (Alam et al., 2021).

Different bin size analysis demonstrated a clear distribution of length parameters, showcasing central tendencies, interquartile ranges, and variability. The robustness of the modeling approach was validated through the consistent estimates of L_{∞} , L_{mat} , L_{max} , L_{opt} , L_{opt_p10} , and L_{opt_m10} across various bin sizes. Deviations in these estimates remained within $\pm 2\%$, and statistical analysis confirmed no significant differences ($p > 0.05$) in parameter values across bin sizes. This robustness is critical for ensuring reliability in length-based stock assessments, as it indicates that the model's performance is unaffected by the choice of bin size. This finding corroborates the work of Hordyk et al. (2015), who demonstrated that length-based methods are robust tools for evaluating fish stock status, even under data-limited scenarios. These findings further support the utility of the aLBI package for fisheries resource management, offering a flexible and reliable tool for estimating key length-based parameters essential for sustainable exploitation and conservation of fish stocks.

4.2.3 Length-based indicators (LBIs)

The findings from length-based indicators (LBIs) estimated using the aLBI package, with bootstrapped confidence intervals, provide critical insights into the fish stock's health. The low percentage of sexually mature fish (12.48%) indicates that approximately 87% of individuals are excluded from the population before reaching maturity, a critical sign of growth overfishing which significantly reduces reproductive capacity and long-term sustainability of the stock. On the hand, the low proportions of optimally sized (9.03%) and mega spawner (5.09%) reflecting the recruitment over-fishing of the fishery. The significant depletion of mega-spawners ($P_{mega} < 20\%$) suggests heavy fishing pressure on larger, older individuals, threatening genetic diversity and resilience (Mora et al., 2009; Barneche et al., 2018). Thus, management strategies should be developed for ensuring the immature individuals can grow and spawn at least once before they are caught and the mega spawners

remain intake (Alam et al., 2022). These strategy will maintain the reproductive success of the population (Froese, 2004). These results align with earlier research showing that overfishing usually results in an overpopulation of young fish, which, under poor management, could cause recruitment overfishing (Beverton & Holt, 1957; Cushing, 1988). Furthermore, ensuring the genetic variety and long-term population health depends on keeping a balanced size distribution (Hixon et al., 2014).

4.2.4 Evaluation of stock status

The stock status probabilities for target spawning biomass (TSB40) and limit spawning biomass (LSB25) across various bin sizes are below their respective target reference points. Moreover, the ratio of length for sexually mature fish to the optimally-sized fish ($L_{mat}/L_{opt} > 0.9$) and the combined percentage of optimally-sized and the mega spawners ($P_{opt} + P_{mega} > 0$) indicate selectivity patterns favoring the capture of smaller and optimally-sized fish while avoiding the larger individuals. Since, there is no strategy for releasing the mega spawners from the catch, this finding also indicates the absence of larger *S. aor* individuals in this Lake. Previous studies (Mia et al., 2024; Bashar et al., 2021) also reported low percentage of mega spawners for *A. coila* and *E. vacha* in this Lake. Given the multi-gear and multi-species nature of the Kaptai Lake fisheries it can be concluded that major portion of mega spawners of different species have already been removed from this Lake by the existing selectivity pattern of the fisheries.

4.3 Recommendations for Management

The findings from the aLBI package emphasize the urgent need for policy interventions to safeguard the reproductive biomass of the fishery. With a high proportion of juveniles in the catch, the optimal target length class (P_{opt}) accounts for only about 10% or less of the total catch composition. Thus, Immediate gear modifications are essential to prevent further stock depletion.

Although existing regulations prohibit the capture of fish ≤ 23 cm in length, this study highlights the inadequacy of such measures in protecting the *S. aor* stock. The analysis suggests that the minimum length limit for this species should be at least 40 cm. Additionally; to conserve older, larger individuals crucial for the reproductive success of the stock, an upper length limit of 49 cm is recommended. However, implementing different size limits for the same gear may not be feasible in resource-limited contexts. Therefore, the study proposes a single lower length limit of 40 cm, ensuring that all fish have the opportunity to mature and spawn at least once in their lifetime. If this regulation is enforced effectively, the fishery can remain sustainable, even with the removal of mega-spawners.

5 Conclusion

The length-based indicators (LBIs) for fisheries sustainability proposed by Froese (2004), have been widely validated and recognized in international literature. Leveraging their proven effectiveness, this study evaluates the stock status of *Sperata aor* from Kaptai Lake, Chittagong, Bangladesh. To provide robust and reliable estimates of these indicators, the study also introduces an R package "aLBI." In addition to estimating LBIs, the package incorporates procedures to address the uncertainties associated with these estimates. It also examines the impact of varying bin sizes on the calculation of indicator values. For the *S. aor* stock, no significant differences were observed across different bin sizes, demonstrating the package's reliability compared to traditional LBI estimation methods. The outputs estimated by the aLBI package reveal that the fishery predominantly targets immature individuals of the species. To safeguard the stock and promote sustainability, this study recommended a lower length limit 40 cm. Enforcing this measure will ensure the sustainability of the fishery in Kaptai Lake.

Author Contributions

A.A.: conceptualization, developing, methodology, data collection, data analysis, visualization, and writing, M.R.S.: writing, reviewing and editing, M.S.A.: conceptualization, methodology, data analysis, writing, reviewing, editing, and supervision. All authors have read and agreed to the published version of the manuscript.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

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Data will be available on request to corresponding author.

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